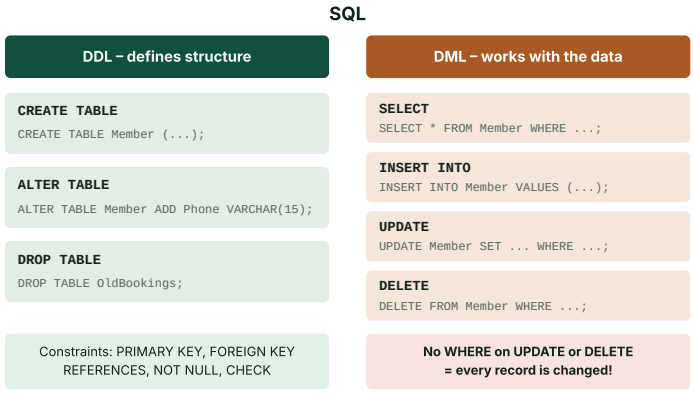
SQL data definition and data manipulation

IS05 FACT FILE

Specification: students should be able to

- Use SQL data definition language (DDL) to create tables with data types, primary keys and foreign keys
- Use SQL data manipulation language (DML) to insert, update and delete records

Key words: SQL, DDL, DML, CREATE TABLE, data type, PRIMARY KEY, FOREIGN KEY, NOT NULL, INSERT INTO, UPDATE, DELETE, constraint



SQL split into data definition language for structure and data manipulation language for data, with an example of each command

DDL and DML

SQL has a data definition language for creating and changing the structure of a database and a data manipulation language for working with the data itself. CREATE TABLE defines a table's fields, data types and constraints; ALTER TABLE changes an existing structure and DROP TABLE removes it. INSERT adds records, UPDATE changes values in existing records, DELETE removes records and SELECT retrieves them. Being declarative, SQL lets the DBMS choose the most efficient way to carry out each command.

Constraints

Constraints written in CREATE TABLE let the DBMS enforce integrity automatically. PRIMARY KEY enforces uniqueness and prevents nulls in the key. FOREIGN KEY with REFERENCES ensures each foreign key matches a primary key in the parent table. NOT NULL acts as a presence check and CHECK enforces a range or other condition. Because constraints live in the schema, every application that uses the database must obey them.

UPDATE and DELETE safely

UPDATE and DELETE apply to every record that satisfies the WHERE clause, so the condition must be precise, normally using the primary key. A missing WHERE affects the whole table. Good practice is to run a SELECT with the same WHERE first to see which rows will be affected, and to use a transaction so the change can be rolled back if it is wrong. Users should only be granted the permissions they need.

T Key terms from the mark schemes

SQL
Structured Query Language: the standard declarative language for defining, querying and manipulating relational databases.

Data definition language (DDL)
The part of SQL used to create and change database structure, e.g. CREATE TABLE, ALTER TABLE, DROP TABLE.

Data manipulation language (DML)
The part of SQL used to retrieve and change data, e.g. SELECT, INSERT, UPDATE, DELETE.

Constraint
A rule defined on a table, such as PRIMARY KEY, FOREIGN KEY, NOT NULL or CHECK, that the DBMS enforces on every change.

Exam focus

Write SQL in the order the syntax expects and use the exact table and field names from the question. Marks are typically given per clause: for example one mark for UPDATE table, one for the SET clause and one for the correct WHERE. Put text values in single quotes and do not quote numbers. In CREATE TABLE questions, choose sensible types and sizes and include PRIMARY KEY and FOREIGN KEY … REFERENCES … where asked. Minor syntax slips are often tolerated if the logic is clear, but missing WHERE clauses are not.

? Check yourself

1. State the difference between DDL and DML, giving one SQL command for each. [2]
2. Write an SQL statement to create a table Member with fields MemberID (6 characters, primary key), Surname (up to 30 characters, required) and JoinDate (date). [4]
3. Write an SQL statement to add a member with ID M0042, surname Corrigan, who joined on 1 September 2026. [2]

Answers: 1. DDL (data definition language) defines / changes the structure of the database, e.g. CREATE TABLE, ALTER TABLE or DROP TABLE (1) / DML (data manipulation language) works with the data itself, e.g. SELECT, INSERT, UPDATE or DELETE (1). 2. Model answer:

```
CREATE TABLE Member (
  MemberID CHAR(6) NOT NULL,
  Surname  VARCHAR(30) NOT NULL,
  JoinDate DATE,
  PRIMARY KEY (MemberID)
);
```

Marks: CREATE TABLE Member with fields in brackets, correct syntax (1) / MemberID CHAR(6) (accept VARCHAR(6)) (1) / Surname VARCHAR(30) NOT NULL (1) / JoinDate DATE and PRIMARY KEY (MemberID) (1). 3. Model answer:

```
INSERT INTO Member (MemberID, Surname, JoinDate)
VALUES ('M0042', 'Corrigan', '2026-09-01');
```

Marks: INSERT INTO Member with field list (accept omitted field list if values are in table order) (1) / VALUES with text and date in quotes in matching order (1). Accept #01/09/2026# (Access) or '01/09/2026'.